Adversarial Manipulation of Explanations in Face Verification

Kshitij Parashar
Human and Identity Centered Machine Learning Project

***Abstract*—Saliency maps are offered alongside face verification decisions so that a human reviewer can inspect the machine's reasoning and catch its mistakes. We ask whether that safeguard can be defeated: can an imperceptible perturbation to one image of a pair leave the decision intact while making the explanation misleading? On ElasticFace-Arc and the full LFW benchmark, an $L_\infty \leq 4/255$ perturbation preserves the decision on 100% of 2,991 accepted genuine pairs and holds the similarity score to within ± 0.026, while relocating 75% of the saliency map's hot-spots and driving map correlation to 0.236. Evaluated with insertion and deletion curves instead, however, the attacked map retains 89% of its faithfulness: the pixels it now highlights still carry the match. Relaxing the score constraint tenfold, or targeting faithfulness directly, closes at most 21.5% of the distance to a random-saliency floor. Two mechanisms set this ceiling: redundancy inside the face, where the attack merely swaps one identity-bearing region for another, and learned invariance outside it, where the model has no gradient to exploit. Judged on correlation alone, this run would have been reported as a successful attack. It is not one, and only a faithfulness-based evaluation makes the difference visible.**

***Index Terms*—face verification, explainable AI, adversarial attacks, saliency maps, faithfulness**

I. Introduction

Face recognition is deployed where the consequences of error are severe: border control, forensic casework, evidence in court. Explainability methods such as xFace [1] and xSSAB [2] are proposed for these settings, on the argument that a saliency map lets a human auditor verify that the system attended to the right evidence.

That argument rests on an unexamined premise: that the explanation is trustworthy in a way the decision is not. A saliency map is another model output computed from the same weights and the same input, and the adversarial fragility of explanations is established for image classifiers [3], [4]. If the same holds for verification, the explanation is not an independent check. It inherits the vulnerability of the system it audits, while appearing more trustworthy because it is legible.

Threat model. A verification system reports *"same person"* and shows a saliency map justifying the decision. An adversary who can modify *one* of the two submitted images adds a perturbation that leaves the decision and the similarity score essentially unchanged while making the explanation point somewhere misleading. The auditor sees a normal decision with a fabricated justification. This is realistic wherever image files are submitted and later reviewed, though not for live camera capture. Note the asymmetry with standard adversarial attacks on biometrics, which aim to flip the decision: here the decision must survive untouched, and the explanation alone is under attack.

Contributions. We measure a faithfulness baseline for four explanation methods against a random-saliency floor (Section III); build an attack that preserves the decision on 100% of pairs while relocating 75% of saliency hot-spots, yet leaves the map 89% as faithful (Section IV); diagnose the resulting ceiling as redundancy inside the face and learned invariance outside it (Section V); and show that correlation and faithfulness are dissociable, so correlation-only evaluation of explanation attacks is systematically optimistic (Section VI).

II. Experimental Setup

We use ElasticFace-Arc [5], an iResNet-100 producing 512-d embeddings, with similarity the cosine of the L_2-normalised embeddings. The data is LFW view-2 [6]: 12,000 images forming 6,000 pairs in 10 folds of 600, each fold being 300 genuine pairs followed by 300 imposter pairs. We verified this layout rather than assuming it, since the labelling determines every downstream number: it separates the two score distributions at Cohen's $d = 8.32$ against $d \approx 0.01$ for two competing hypotheses, and holds in all ten folds independently.

Each pair is split into a *probe*, the single image an attack may modify, and a *reference*, which is never touched, by a seeded per-pair coin flip so that the attacked image is not systematically image 0. Rather than fixing a threshold by hand we derive it: the Equal Error Rate is 0.300% at $\tau = 0.2136$, and every decision-preservation result is reported additionally at $\mathrm{FMR} = 10^{-1}, 10^{-2}, 10^{-3}$, so that no conclusion rests on a single operating point. Under the standard LFW 10-fold protocol the pipeline reaches $99.800\% \pm 0.246\%$, matching the published ElasticFace-Arc figure; that can only come out right if the weights, the preprocessing and the pair labelling are all correct, so it validates everything downstream. Of the 3,000 genuine pairs, 2,991 are accepted at τ and form the attack population.

As a warm-up we confirmed that the pipeline can *break* a decision before asking it to preserve one: FGSM [7] and I-FGSM [8] in both directions reach 100% success at $4/255$ with zero failures (Appendix A).

III. Are the Explanations Faithful to Begin With?

A post-attack faithfulness number is meaningless without two reference points: how good the explanation was initially, and what "no information" scores. We measure both first.

TABLE I
FAITHFULNESS BEFORE ANY ATTACK

method	deletion AUC ↓	insertion AUC ↑	gap ↑
gradient	0.1793	0.5597	**0.3803**
xFace	0.1772	0.5287	0.3516
xSSAB (combined)	0.1971	0.5319	0.3349
random (floor)	0.3698	0.4806	0.1108

Genuine pairs, $n = 3{,}000$. The random row is the floor: what a structureless map scores. It is the scale against which every later number is read.

We use insertion and deletion curves [9], adapted to verification. Both perturb the probe and track cosine similarity to the untouched reference, in the order the saliency map ranks pixels. *Deletion* replaces the most salient pixels with mid-grey, so a faithful map destroys the match early and low AUC is good. *Insertion* restores them over a blurred probe, so a faithful map rebuilds the match early and high AUC is good. We evaluate four maps over all 6,000 pairs: the squared input-gradient [10], our implementation of xSSAB [2], xFace occlusion [1], and uniform noise as the control that supplies the floor.

Every real method separates decisively from noise (Table I): gradient beats random at Cohen's $d = 3.44$, an order of magnitude larger than any difference *between* methods ($d = 0.16$ to 1.14).

Two findings carry forward. Insertion is a weak discriminator here, since random insertion already reaches 0.481: restoring any half of the pixels over a blurred baseline recovers most of the similarity. Headroom is 0.08 for insertion against 0.19 for deletion, so attacks should be judged on deletion. And imposter pairs are degenerate for this metric, since at a base similarity of 0.003 there is almost no match evidence to localise, which is why the attack targets accepted genuine pairs.

IV. THE EXPLANATION ATTACK

The attack minimises a single loss, one gradient-descent step per iteration:

$$\mathcal{L} = \underbrace{\lambda_{\text{pred}}\,\text{relu}\big(|s - s_0| - \text{tol}\big)}_{\text{keep the score where it was}} + \underbrace{\lambda_{\text{expl}}\,\rho\big(M_{\text{att}}, M_0\big)}_{\text{move the explanation}} \quad (1)$$

where s_0 and M_0 are the clean similarity and the clean saliency map and ρ is Pearson correlation.

Two design choices matter. The decision term is two-sided, penalising the score for moving in *either* direction rather than merely for falling below a floor, since a one-sided floor lets the attack push similarity upward and weakens the claim. The explanation term uses correlation rather than MSE because correlation is scale-invariant: rescaling a map produces a large MSE while leaving the pattern a reviewer actually sees unchanged. After each step the perturbation is projected into the ε-ball and shrunk per sample if the score leaves the band. We run 2,991 accepted genuine pairs at $\varepsilon = 4/255$, 40 steps, $\text{tol} = 0.05$.

The decision survives. Mean $\Delta s = +0.0110$ and mean $|\Delta s| = 0.0261$, with 95% of pairs below 0.0474. The decision is preserved on 100% of pairs at all four operating points, at a mean L_∞ of $2.67/255$. The score does not merely stay on the correct side of a line; it stays where it was.

The map moves a great deal. Pearson correlation with the original map falls to 0.236 on average (median 0.128), with 75.7% of pairs below 0.3, and top-10% hot-spot overlap is 0.253, so three quarters of the originally hottest pixels relocate. By any map-comparison metric, the reviewer is shown a different explanation of a decision that has not moved (Fig. 1).

But the map does not become wrong. Measured by deletion AUC instead (Table II), the rise is overwhelmingly significant ($p \approx 10^{-221}$ at $n = 2{,}991$) but small: the optimised gradient map travels only 10.9% of the distance to the random floor, so it retains 89% of its faithfulness. The effect is a real mechanism rather than noise, since per pair the map movement predicts the faithfulness lost ($r = -0.347$, $p \approx 10^{-85}$) and the 200 most-disrupted pairs shift nearly three times the average. It is simply bounded.

The attack therefore scrambles what the map looks like without substantially destroying what it knows. Table II adds a second limitation: correlation after the attack rises monotonically with distance from the optimised map, from 0.236 for the gradient to 0.720 for the gradient-free xFace. The attack compromises a *family* of explanations, not the model's reasoning, so auditing with an occlusion-based tool is a cheap and concrete defence.

V. WHY THE ATTACK HAS A CEILING

Correlation collapses while faithfulness does not, which leaves a gap to explain. Two hypotheses are testable. **H1**, the budget: the ± 0.05 band shrank the perturbation on 49.9% of pairs, so perhaps the attack was never allowed enough room. **H2**, the objective: the attack minimises correlation, but faithfulness is a different quantity, and a face is redundant. Eyes, nose bridge, mouth and jawline all carry identity, so the attack can lower correlation by moving the ranking onto a different set of genuinely informative pixels.

H1 is real but not sufficient. Widening the score band (Table III) buys genuine disruption, but the marginal returns saturate ($+8.5$, $+6.7$, $+2.5$ points), and at the widest band the constraint is effectively off: only 2% of pairs are shrunk, mean L_∞ is 3.95 of a 4.00 budget, and correlation is already floored at 0.065. Even there the attack covers only 21.5% of the distance to the random floor.

H2 is confirmed. At $\text{tol} = 0.05$, off-face saliency mass rises only $0.239 \to 0.275$, hot-spots remaining on the face fall only $76.9\% \to 72.0\%$, and the centre of mass moves a median of 7.84 px on a 112 px image. The mass stayed on the face and moved *within* it, swapping one identity-bearing region for another, which is why correlation collapsed while faithfulness did not.

Attacking faithfulness directly does better, and still hits a wall. Replacing the correlation term in (1) with the fraction of saliency mass falling outside the face lifts the attack from 12.4% to 15.4% of the way to the floor ($1.24\times$, $p = 0.014$), which demonstrates rather than argues that correlation and faithfulness are dissociable. But the tighter region makes the

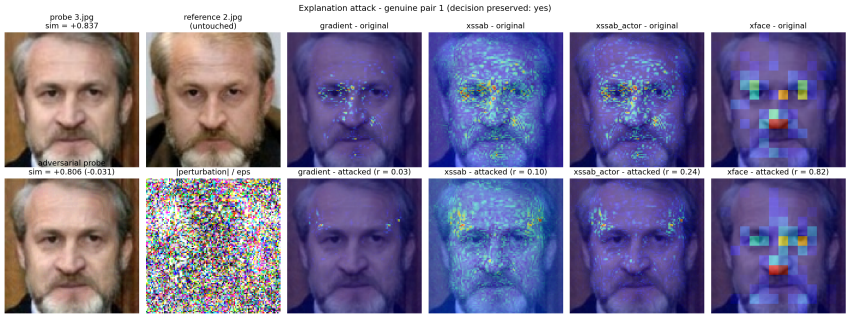


Fig. 1. The attack on genuine pair 1. Probe `3.jpg` is modified, reference `2.jpg` is not; the two probe images are indistinguishable and the score moves $0.837 \rightarrow 0.806$. Reading the maps left to right shows the transfer staircase within a single pair: the targeted gradient map is destroyed ($r = 0.03$), xSSAB partly ($r = 0.10$), the actor variant less ($r = 0.24$), and the gradient-free xFace map is essentially untouched ($r = 0.82$).

TABLE II
EXPLANATION METRICS BEFORE AND AFTER THE ATTACK

	map similarity (lower = more disrupted)		faithfulness, deletion AUC (lower = more faithful)			
method	Pearson r	top-10% IoU	before	after	to floor	retained
gradient (optimised)	**0.236**	0.253	0.1799	0.2007	10.9%	**89%**
xSSAB (combined)	0.341	0.225	0.1976	0.2232	14.9%	85%
xSSAB (actor)	0.458	0.251	0.1801	0.2005	10.8%	89%
xFace (occlusion)	**0.720**	0.443	0.1777	0.1952	9.1%	91%

The 2,991 accepted genuine pairs. The attack optimises the gradient map only, so the map-similarity columns read as a transfer staircase. *To floor* is $\Delta\mathrm{del}/(0.3708 - \mathrm{del}_{\mathrm{before}})$, the fraction of the distance from the unattacked value to the random-saliency floor, which is 0.3708 on this population; *retained* is its complement.

attack *harder*. These are ArcFace-aligned crops [11] in which the face fills the frame; under a background ellipse covering 37.8% of the image, the unattacked explanation places only 5.3% of its mass there, a concentration of 0.14 relative to area. The attack quadruples that to 0.215, a concentration of 0.57, well short of the no-preference level of 1.0. An ArcFace-trained network is invariant to background: the input gradient there is near zero, so a $\pm 4/255$ perturbation cannot manufacture sensitivity where the model has none. The attack is fighting a learned invariance, not an optimisation difficulty.

VI. DISCUSSION

Two mechanisms jointly explain the ceiling. Inside the face there is *redundancy*: multiple regions carry identity, so relocating the ranking among them preserves faithfulness while destroying correlation. Outside the face there is *invariance*: the model has no gradient in background, so mass cannot be pushed there. Together they say that the saliency map's support region is a property of the model, so an imperceptible perturbation can shuffle mass within that support but cannot move the support itself. This is a positive claim about robustness rather than a failed attack, and it predicts a structural ceiling on any model combining background invariance with a redundant signal.

The methodological consequence is the more important result. With correlation as the only metric, this study would have reported a successful attack at $r = 0.236$. Insertion and deletion separate *"the explanation changed"* from *"the explanation became misleading"*, and show that the first happened while the second largely did not. Correlation-based evaluation of explanation attacks is therefore likely to be systematically optimistic: as with sanity checks for saliency maps [12], a metric measuring *change* is not measuring *correctness*.

One caveat runs the other way. Insertion and deletion ask whether the highlighted pixels are informative, not whether the model relied on them most, so telling a reviewer *"the cheek drove this match"* when the model used the eyes misleads even if the cheek carries identity evidence. Both statements stand: the attack changes what a human is told, while largely failing to make the explanation wrong.

Limitations. The attack is white-box and digital only, with no cross-model transfer and no JPEG or print-scan robustness; LFW is near-saturated for ElasticFace-Arc; xSSAB is our implementation rather than the official code; insertion and deletion perturb inputs off-distribution, a confound the random control bounds but does not remove; and only accepted genuine pairs were attacked. The clearest next step is a multi-method attack optimising against gradient, xSSAB and a differentiable occlusion surrogate at once, testing whether the occlusion defence survives.

VII. CONCLUSION

An imperceptible perturbation to one image of a pair holds the decision and the score fixed while rewriting three quarters of the saliency map's hot-spots. By map correlation, the standard in this literature, that is a successful attack; by faithfulness it is not. The saliency map's support is a property of the model rather than of the input, and an imperceptible perturbation cannot move it.

References

[1] M. Knoche, T. Teepe, S. Hörmann, and G. Rigoll, "Explainable model-agnostic similarity and confidence in face verification," in *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2023, pp. 711–718.

[2] M. Huber, A. T. Luu, P. Terhörst, and N. Damer, "Efficient explainable face verification based on similarity score argument backpropagation," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 4736–4745, arXiv:2304.13409.

[3] A.-K. Dombrowski, M. Alber, C. J. Anders, M. Ackermann, K.-R. Müller, and P. Kessel, "Explanations can be manipulated and geometry is to blame," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[4] J. Heo, S. Joo, and T. Moon, "Fooling neural network interpretations via adversarial model manipulation," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[5] F. Boutros, N. Damer, F. Kirchbuchner, and A. Kuijper, "ElasticFace: Elastic margin loss for deep face recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2022, pp. 1578–1587, arXiv:2109.09416.

[6] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," University of Massachusetts, Amherst, Tech. Rep. 07-49, 2007.

[7] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *International Conference on Learning Representations (ICLR)*, 2015.

[8] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *International Conference on Learning Representations (ICLR) Workshop*, 2017.

[9] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in *British Machine Vision Conference (BMVC)*, 2018.

[10] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *International Conference on Learning Representations (ICLR) Workshop*, 2014.

[11] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699.

[12] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Appendix A
Protocol and Decision Attacks

Before claiming that an attack *preserves* a decision, the pipeline must be able to *break* one on purpose. Running FGSM and I-FGSM in both directions over all 6,000 pairs, at $4/255$ with I-FGSM both directions reach 100% with zero failures, with mean Δs of -1.116 and $+0.691$ respectively. Success holds at approximately 100% at every operating point from $\mathrm{FMR} = 10^{-1}$ to 10^{-3}.

An asymmetry appears at low budget: impersonation succeeds on 62.4% of pairs at $1/255$ against only 6.7% for obfuscation. Two competing effects produce this. Obfuscation displaces the score *more* per unit of perturbation ($1.18\times$ at $1/255$, rising to $1.73\times$ at $8/255$), since pushing an embedding away from a point is easier than steering it toward one. But obfuscation also has $2.41\times$ further to travel, because $\tau = 0.2136$ sits close to the imposter mean (gap 0.210) and far from the genuine mean (gap 0.506). The second effect dominates. Attack success rates are therefore as much a statement about where the operating threshold sits as about the attack itself, which is why the threshold had to be derived before any of this could be interpreted.

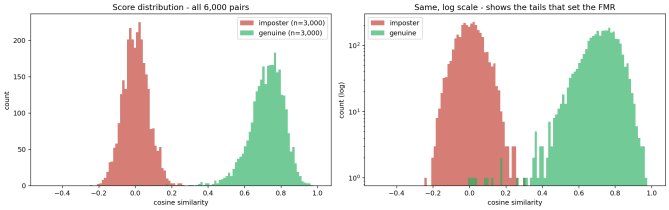


Fig. 2. Genuine (green) and imposter (red) cosine-similarity distributions over all 6,000 pairs, linear and log scale. The log panel is the one that matters: with 3,000 imposter scores the tail is resolved well enough to estimate a false match rate, which the commonly used 900-pair subset could not support.

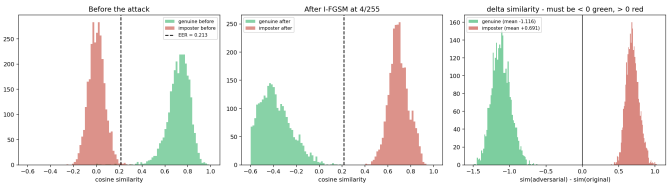


Fig. 3. I-FGSM at $4/255$. The two populations swap sides of the threshold completely. The right panel is the direction check: genuine Δs must be negative and imposter positive, and the two distributions do not overlap zero.

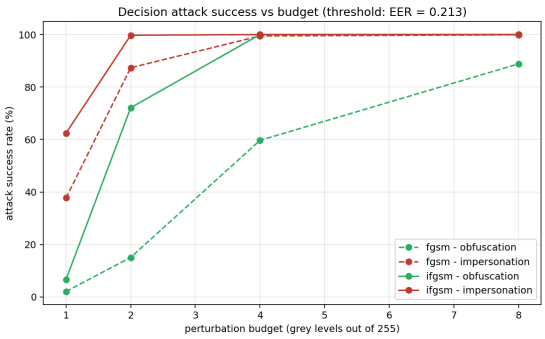


Fig. 4. Success rate against budget at the EER threshold. Impersonation (red) dominates at small budgets despite obfuscation displacing the score further per unit of ε, because the threshold sits closer to the imposter distribution. Both saturate at $4/255$.

Appendix B
Explanation Faithfulness

An aside on xSSAB. Measuring both variants shows that the actor map alone scores deletion 0.1801 and insertion 0.5613, identical to the raw input-gradient map, while the combined actor minus counteractor map is clearly worse (0.1976 / 0.5334). Any ranking that places xSSAB last is an artefact of scoring the combined map; the actor variant is the one to use.

Appendix C
Attack Diagnostics

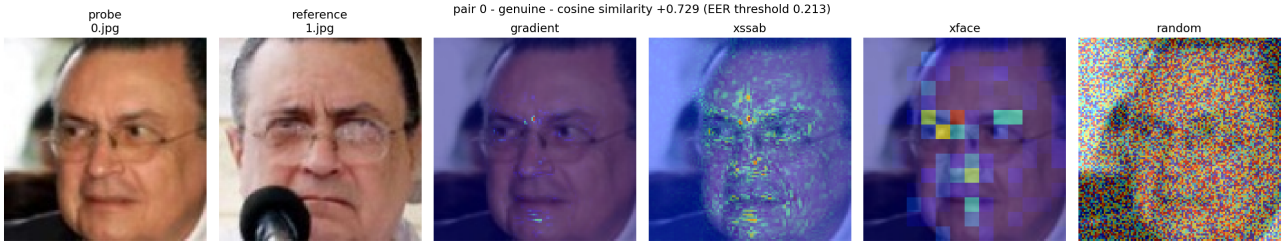


Fig. 5. The four maps on genuine pair 0 (similarity $+0.729$). Gradient is pixel-sharp and sparse, xSSAB denser and more semantic, xFace blocky because it is grid-based, and the random control structureless. All three real methods concentrate on the face interior.

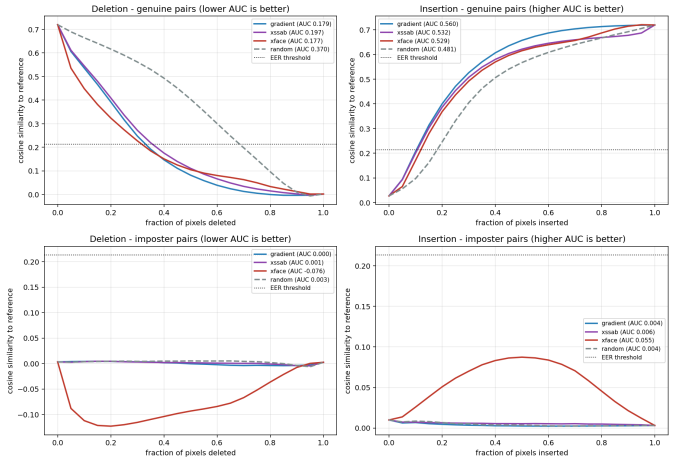


Fig. 6. Mean insertion and deletion curves over all 6,000 pairs, genuine (top) and imposter (bottom). For genuine pairs every real method drops far faster than random under deletion and rises faster under insertion. The imposter row shows the degeneracy noted in Section III: with almost no match evidence to localise, all curves collapse toward zero. xFace is the informative exception, its deletion AUC going negative (-0.076) because it correctly identifies the few regions supplying positive evidence.

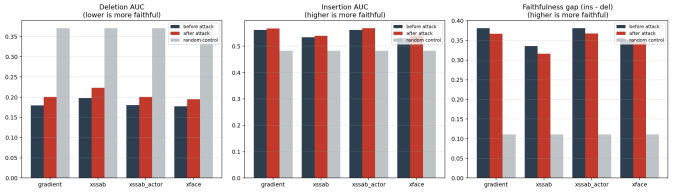


Fig. 7. Faithfulness before (dark) and after (red) the attack, against the random-saliency control (grey). The attacked bars barely move from the unattacked ones and remain far from the control: the explanation changed, but it did not become uninformative. Insertion (middle) moves so little that it is visually flat, consistent with its narrow headroom. These are the numbers of Table II in graphical form.

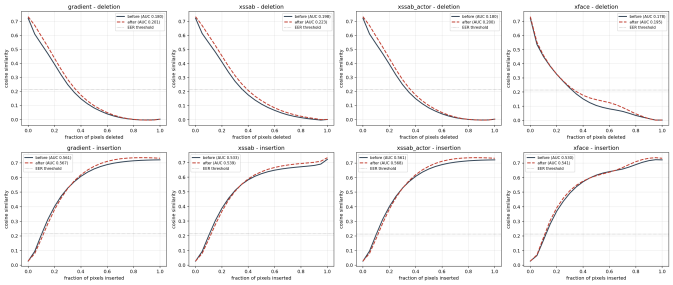


Fig. 8. The main result as curves rather than areas. The attacked deletion curves (dashed red) sit only slightly above the unattacked ones (solid), and the gap closes entirely past roughly 60% of pixels deleted. A successful attack would have lifted the dashed curve toward a straight diagonal.

TABLE III
SCORE-BAND SWEEP

tol	mean $\lvert\Delta s\rvert$	shrunk	Pearson r	del. AUC	to floor
0.02	0.0118	88%	0.479	0.1983	3.9%
0.05	0.0260	44%	0.220	0.2135	12.4%
0.10	0.0462	14%	0.109	0.2254	19.1%
0.20	0.0606	**2%**	**0.065**	0.2298	**21.5%**

A 300-pair subset with its own unattacked baseline (deletion AUC 0.1915) and floor (0.3698), so *to floor* is not directly comparable with Table II. The $\mathrm{tol} = 0.05$ row is the configuration used in Section IV. *Shrunk* is the fraction of pairs whose perturbation the band pulled back below the budget.

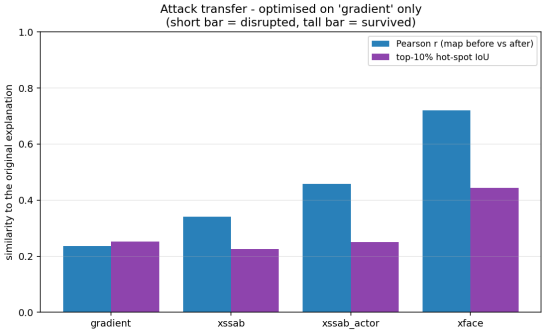


Fig. 9. Transfer. Both map-similarity metrics rise monotonically with distance from the optimised map, and xFace, which uses no gradients, retains most of its explanation under both.

TABLE IV
ATTACKING FAITHFULNESS DIRECTLY

off-face region	area	mass	concentration	to floor	vs. decorrelation
border frame	67.3%	$0.239 \rightarrow 0.614$	$0.36 \rightarrow 0.91$	19.9%	$1.61\times$, $p = 5.6\times10^{-9}$
ellipse (background)	37.8%	$0.053 \rightarrow 0.215$	$0.14 \rightarrow 0.57$	15.4%	$1.24\times$, $p = 1.4\times10^{-2}$

The same 300 pairs and score band as Table III. Mass and concentration are given as before $\rightarrow$ after; concentration is mass share divided by area share, so 1.0 means no spatial preference. The decorrelation objective reaches 12.4% of the way to the floor on this subset.

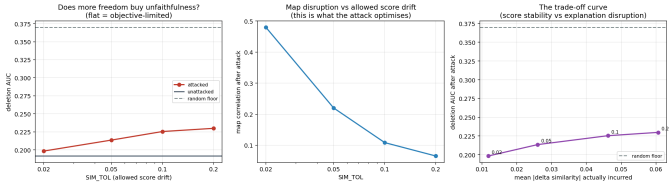


Fig. 10. Score-band sweep. Left: deletion AUC against the allowed score drift, with the unattacked level and the random floor marked; the curve rises but flattens and never approaches the floor. Middle: the quantity the attack actually optimises falls steadily to 0.065, so it is not the correlation objective failing to converge. Right: the trade-off curve between score stability and explanation disruption.

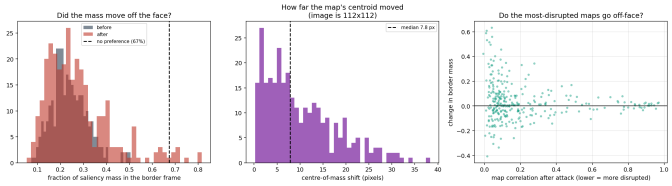


Fig. 11. Where the saliency mass went. Left: the distribution of off-face mass barely shifts under attack and stays far below the no-preference line. Middle: the centre of mass moves a median of 7.8 px on a 112 px image. Right: even the most-disrupted maps do not systematically push mass off the face. Measured under the wider border-frame definition.

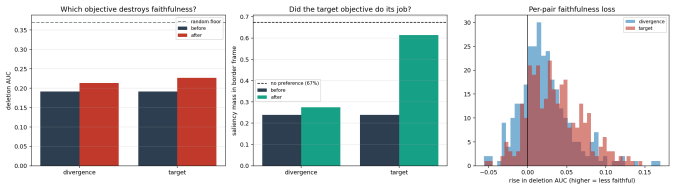


Fig. 12. Target objective against decorrelation. Left: both objectives lift the deletion AUC, the target objective further, but neither reaches the random floor. Middle: the target objective does succeed at its own task, as off-face saliency mass rises sharply. Right: the per-pair distribution of faithfulness lost is shifted right for the target objective. Border-frame definition; the ellipse result is in Table IV.



Fig. 13. The two objectives on the same pair. Under `divergence` (middle row) the saliency relocates from the eyes to the mouth and chin, still on the face. Under `target` (bottom row) a visible cluster appears in the upper-right background and off-face mass more than doubles. Both leave the photograph unchanged to the eye and the score within the band ($+0.729 \rightarrow +0.748$ and $+0.692$). This is redundancy and invariance in one picture: the attack can move mass, but what it moves it *onto* still has to be somewhere the model looks.